

Dashboard Tiles Reference — Meaning and Data Sources

ADF Analyzer v10.1 Dashboard | Last updated: 2025-11-08 | **Production Ready Edition**

Overview

This document describes **every metric tile** shown by the dashboard in `armv10/adf_dashboard.py`:

-  **What the tile means** (business purpose)
-  **Data source sheets** and fallback logic
-  **Calculations and formatting** applied
-  **Configuration controls** in `enhancement_config.json`

 **Quick Access:** View this documentation directly in the dashboard sidebar under " Documentation"

Top-Row Metric Tiles (Enhanced Metrics)

Rendered by `render_enhanced_metrics()` function

1. Pipelines

- Meaning: Number of pipeline resources discovered in the factory.
- Data source / fallback logic:
 - Primary: `Summary` sheet metric `Pipelines` when present (via `get_summary_metric`).
 - Fallbacks (in order): count rows in any of these sheets if `Summary` missing: `ImpactAnalysis`, `PipelineAnalysis`, `Pipeline_Analysis`, `Pipelines`.
- Type: integer count.

2. DataFlows

- Meaning: Number of Data Flow resources discovered.
- Data source / fallback logic:
 - Primary: `Summary` sheet `DataFlows`.
 - Fallbacks: `DataFlows`, `DataFlowLineage`, `DataFlow_Summary` sheets (row count or summary metric).

3. Datasets

- Meaning: Number of dataset resources discovered.
- Data source: `Summary` sheet `Datasets` or fallback to the `Datasets` sheet row count.

4. Triggers

- Meaning: Number of triggers (scheduled/evented) discovered.
- Data source / fallback logic (explicit):

- Primary: **Summary** sheet **Triggers** when present.
- Preferred sheet: **Triggers** (one row per trigger) — use this count when present.
- Fallback: **TriggerDetails** — this sheet contains trigger occurrence/usage rows and can include multiple rows per trigger. When **Triggers** is absent the dashboard/validator will count unique trigger names from **TriggerDetails** (if a sensible name column like **Trigger** exists); otherwise it falls back to the raw row count.
- Rationale: some analyzer outputs include a separate **Triggers** sheet listing unique triggers (canonical). **TriggerDetails** is richer (pipeline associations, schedules) but counts of its rows do not equal unique trigger definitions.

5. Dependencies

- Meaning: Total number of dependency edges found (activity/pipeline triggers/dependsOn relationships).
- Data source / fallback logic:
 - **Summary** sheet metric **Total Dependencies** when present.
 - Fallback sheets used to build dependency counts: **ActivityExecutionOrder**, **DataLineage**, **Pipeline_Pipeline**, **Pipeline_DataFlow**, **TriggerDetails**.

6. Health

- Meaning: Factory-level health score (0–100) based chiefly on orphaned pipelines as a fraction of pipelines.
- Data source / calculation:
 - **pipelines** (see Pipelines) and **orphaned** (see Orphaned tile) are used.
 - Formula (canonical):
 - If pipelines > 0:

$$\text{health_score} = \text{int}((1 - \text{orphaned} / \text{pipelines}) * 100)$$
 - Else: health_score = 100
 - The gauge (**render_health_gauge**) maps the numeric score into statuses:
 -  =90 → Excellent (green)
 - 75–89 → Good (blue)
 - 60–74 → Fair (yellow)
 - <60 → Needs Attention (red)
- Controlled by: **enhancement_config.json** → **advanced_dashboard.health_score** (true/false)

7. Orphaned

- Meaning: Count of orphaned pipelines (resources with no inbound references/usage).
- Data source:
 - **OrphanedPipelines** or **Orphaned_Pipelines** sheets (row count) or Summary metric **Orphaned Pipelines**.
- Formatting: When orphaned > 0 the tile uses a warning gradient and shows a warning emoji.

Secondary row: Source/Target and Static vs Dynamic metrics

Also rendered by `render_enhanced_metrics()`.

1. Source Datasets

- Meaning: Unique count of source datasets (distinct dataset names that appear as sources in lineage).
- Data source: `DataLineage` sheet / dataframe column `Source` (unique non-null count).

2. Target Datasets

- Meaning: Unique count of sink/target datasets.
- Data source: `DataLineage` sheet / dataframe column `Sink` (unique non-null count).

3. Static Sources / Static Targets

- Meaning: Count of source/sink entries that appear to be concrete/static dataset identifiers.
- Detection rule: `SourceTable/SinkTable` values that do NOT contain parameterization patterns such as `@dataset`, `@{`, `pipeline()`, `activity()`.
- Data source: `DataLineage` sheet columns `SourceTable` and `SinkTable`.

4. Dynamic Sources / Dynamic Targets

- Meaning: Count of parameterized or templated dataset references (suggesting runtime variation).
- Detection rule: `SourceTable/SinkTable` values that contain `@dataset`, `@{`, `pipeline()` or `activity()`.
- Data source: `DataLineage` sheet.

Impact level and severity tiles

These appear in impact analysis and the "impact level metrics" row.

- CRITICAL / HIGH / MEDIUM / LOW counts
 - Meaning: Number of pipelines (or items) flagged with that impact level.
 - Data source: `ImpactAnalysis` or `impact/Impact` column in pipeline analysis dataframes. If missing, the analyzer defaults some impacts to `LOW` for display.
 - Ordering and color mapping are controlled by `impact_order = {'CRITICAL':0, 'HIGH':1, 'MEDIUM':2, 'LOW':3}` and colors defined in code (CRITICAL red, HIGH orange, MEDIUM yellow, LOW green).

Overview & utility tiles (summary area)

Across the UI there are a few extra tiles and metrics:

- Sheets Loaded
 - Meaning: Number of sheets read from the analysis workbook.
 - Data source: `metadata.get('sheets', [])` length.

- Total Records
 - Meaning: Sum of rows across data sheets that contribute to the analysis.
 - Data source: calculated `total_records` aggregated during metadata extraction.
 - Total Lineage Records / Unique Sources / Unique Sinks
 - Meaning: Lineage-specific metrics derived from `DataLineage` dataframe (counts of records, unique sources, unique sinks).
 - Activity-specific counts (e.g., Copy Activities)
 - Meaning: Counts of activities of a particular type discovered during parsing.
 - Data source: `ActivityCount` or similar sheets; the dashboard aggregates by `ActivityType`.
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Where tile data comes from (summary)

- Preferred single source: `Summary` sheet (if present) — many top-level metrics are read from named `Metric` rows in this sheet.
 - If `Summary` is missing or incomplete, the dashboard uses sheet-specific fallbacks (counts from canonical sheet names). Example fallback mapping is implemented in `get_count_with_fallback()` in `adf_dashboard.py`.
 - Lineage and source/target classification use `DataLineage`.
 - Circular detection / orphan counts come from `CircularDependencies` and `Orphaned*` sheets generated by the analyzer.
 - Impact levels come from `ImpactAnalysis` / `impact` column in pipeline analysis results.
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Where to change tiles or formulas

- `adf_dashboard.py` is the place to change tile labels, HTML/CSS, or visual mapping.
 - `adf_analyzer_v10_excel_enhancements.py` and analyzer modules are the place to change detection logic (e.g., how orphans or circular deps are computed) or to change scoring/impact assignment.
 - `enhancement_config.json` toggles features (see `advanced_dashboard` / `health_score`, `activity_distribution`, `network_stats`, etc.).
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Notes & assumptions

- The UI code includes defensive fallbacks for missing sheets: blank or missing summary metrics do not crash the UI; counts default to 0 and are derived from fallback sheets.
- The dashboard treats circular dependencies as CRITICAL (must-fix). See `CircularDependencies` sheet and `LOGIC.md` for scoring impacts.
- If you want image icons instead of emojis, we can replace the emoji glyphs with embedded SVGs or data-URI images for crisp cross-platform rendering.

If you'd like, I can add a short link from the dashboard homepage to this [TILES.md](#) and embed a little legend tooltip for each tile.